COMSATS University Islamabad, Lahore Campus

Midterm Examination – FALL 2023

Course Title:	Web Technologies			Course Code:	CSC336	Credit Hours:	4(3,1)
Instructor/s:	Muhammad Usman Akram, Asif Shahzad, Ali Hassan			Program Name:	BSCS, BSSE		
Semester:		Batch:	Section:		Date:	02-11-2023	
Time Allowed:	1.5 Hours			Maximum Marks:		50	
Student's Name:				Regi No.			

Important Instructions/ Guidelines:

- Please be to the point. No need to add unnecessary details.
- Please make sure to review all the answers before submitting the paper.
- Please spend some time understanding the question and plan for the answer.
- Minor mistakes of syntax are not a big issue.

Question 1 [CLO-1] **[5+5+5+5]**

Suppose you have following HTML code inside a document:

```
<div id="goToTop"> <a href="">Go To Top</a> </div>
```

a) Use CSS to place this div at the bottom left of the screen.

b) Suppose you have a div with id "menu" placed at the top of the page. Modify the hyperlink in above code such that if it is clicked it should scroll the page to top.

c) Create a web page that showcases two images. Implement a visual effect where one of the images appears to overlap the other. Utilize the z-index CSS property to control the stacking order and make one image take precedence over the other.

d) Suppose you have a web page that displays a list of subjects for the current semester. Each subject should be represented as a hyperlink. When a user hovers over a hyperlink, it should change its appearance using CSS pseudo-classes. Additionally, when a hyperlink is clicked, it should open a respective subject page (empty pages are acceptable). Provide both the HTML and CSS code to achieve this.

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Question 2 [CLO-3] **[5+5]**

Suppose you have following html code and jquery is already loaded in this document

```
<label for="operand1">Operand 1:</label> <input type="text" id="operand1">
<label for="operand2">Operand 2:</label> <input type="text" id="operand2">
<button id="calculate">Calculate</button>
<div id="result"> </div>
```

Write a JS/JQuery code

a) define handler and bind it to click event of the button
b) when button is clicked, the handler should submit the data to an api endpoint at localhost:3000/api/calculate using post method and add the result in div with id "result".

Question 3 [CLO-3] **[10]**

Suppose you have a simple form that collects the user's name, email, and message. Write a JavaScript function **validateForm()** that checks if any of the fields are empty and shows an alert on the screen. The form's **onsubmit** attribute is set to call the **validateForm()** function. If the function returns **false** (indicating validation failure), the form submission is prevented. If it returns **true**, the form is allowed to submit.

```
<form id="myForm" onsubmit="return validateForm();">
    <label for="name">Name:</label>
    <input type="text" id="name" name="name" required><br><br>

    <label for="email">Email:</label>
    <input type="email" id="email" name="email" required><br><br>

    <label for="message">Message:</label>
    <textarea id="message" name="message"  required></textarea><br><br>

    <input type="submit" value="Submit">
</form>
```

Question 4 [CLO-2] **[5+5]**

Consider below code and write updated version of code along with required CSS that shall fulfill below requirements. you can use Bootstrap grid system related classes or write your own CSS. If you decide to write your own CSS, work for a 500px breakpoint.

```
<div class="row">
<div class="col">A</div>
<div class="col">B</div>
<div class="col">C</div>
</div>
```

a. In large and above screen sizes, all three columns shall take equal width.
b. In medium and below screen sizes, first two columns should take equal width and should come in a row. and the 3rd column should take 100% width and shall shift to a new row.

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